

NEURAL ARCHITECTURE ROADMAP PROMPT

Project Goal

Analyze, brainstorm and build a roadmap, outline, and timeline for achieving the following in 15 years or less.

Core Question

Now let's talk about algorithm complexity, constraints, and requirements. What are the requirements of an algorithm, or an artificial intelligence model that has to, at the bare minimum be able to handle this vast amount of data, and represent an OMNICELLULAR ARCHITECTURE of its mind which is far more advanced to handle running these simulations? Where are we now in comparison to this? How do we get there? What will it take? How expensive would it be to undertake and achieve this? What would find it? In planning, designing, and implementing 1AI, and -1AI, where would this be in the development process and why? What problems arise? What pushback will there be?

Research Findings

To even approximate an “omnicellular” simulator of 2,000 human bodies, you need multiple layers of scaling: data, model capacity, compute, and architecture. Today's systems are many orders of magnitude short of that, but we do have a roadmap for getting partway there.[1][2][3]

1. Bare-minimum technical requirements

For an algorithm/model that can even coherently deal with something on the order of 10^{16} – 10^{17} interacting entities (your 2,000-adult, many-trillions-of-cells mental picture), you'd minimally need:

Huge state space and structured representation

- Some way to represent billions–trillions of entities in compressed form (e.g., multi-scale abstractions: organs → tissues → cell types → effective dynamics), not literally 1 parameter per cell.[4][5]
- Hierarchical or modular architecture where each module handles a scale (molecular, cellular, tissue, system, agent).

Enormous model capacity and memory bandwidth

- Frontier LLMs are in the 100B+ parameter regime; “brainlike” or whole-organism models would almost certainly require effective capacities in the 10^{12} – 10^{15} parameter equivalent range, even if achieved via sparsity/MoE.[2][6][7]
- Fast access to a huge external memory or world model to store and retrieve structured state.

Massive parallel compute and efficient simulation kernels

- Something like exaflop-class hardware just to simulate a human brain in real time is a common back-of-the-envelope.[5]
- Multi-organism or multi-body simulations at cellular resolution would need far beyond that unless they are radically abstracted.

Scaling-law-aware training pipeline

- You need to obey empirical scaling laws: performance improves predictably with more parameters, data, and compute following power laws.[1][8][3]
- Compute-optimal scaling (Chinchilla-style) requires growing model size and data together, not just inflating parameters.[9][6][10]

Data that actually defines the target world

- For “omnicellular” cognition, that means large-scale biological, physical, and behavioral datasets, not just text.
 - Whole-brain/whole-organism datasets are currently feasible only for tiny animals, at huge cost.[4]
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2. Where we are now vs that goal

Today’s frontier models:

Operate at much coarser granularity

- LLMs treat the world as tokens and latent vectors, not cells or neurons. They excel at abstract pattern prediction, not microphysical or cellular simulation.[1][2][3]

Scale, but nowhere near “omnicellular”

- We have reliable scaling laws for models up to hundreds of billions of parameters, trained on trillions of tokens.[8][9][3]
- Even exascale-class AI supercomputers are sized around training these models, not simulating many detailed bodies in real time.[9][11][10]

Brain/emulation work is early-stage and tiny

- Whole-brain emulation discussions still talk about sub-million-neuron insect or fish models as near-term, not full mammalian brains.[4]

- Reasonable optimistic estimates put convincing mouse-brain emulation at about a billion dollars in the 2030s, and first human-brain emulations costing tens of billions in the 2040s+ if things go well.[4][12]

So: We can build powerful symbolic/abstract “minds” that reason over text and high-level features; we are nowhere near simulating 2,000 full cellular bodies in detail, and we don’t need to for intelligence.

3. How we might get closer (architecturally)

For something like your OMNICELLULAR ARCHITECTURE, a plausible progression looks like:

Step 1 – Multi-scale world models

- Build agents that maintain explicit multi-scale representations: macroscopic physics, organ-level physiology, micro-level approximations when needed.
- Use learned surrogates (neural PDE solvers, learned biophysical models) instead of direct micro-simulation.

Step 2 – Modular, sparsely activated architectures

- Mixture-of-experts and other sparse models let you have trillion+ potential parameters with only a small subset active per token or per “region.”[6][7]
- Map modules to “organs” or “systems” that can be updated semi-independently but still communicate.

Step 3 – Hybrid symbolic–neural control

- Use symbolic planners or differentiable programs to orchestrate high-level tasks; neural modules handle low-level pattern recognition and approximate dynamics.
- This is necessary to avoid combinatorial blowup when reasoning about billions of entities.

Step 4 – Data and simulators as training teachers

- Instead of training directly on raw cell-level logs, distill knowledge from high-fidelity simulators and experiments into compressed policies/world models.
- Similar to how physics-informed networks approximate PDE solvers but run orders of magnitude faster.

This path lets you approximate an “omnicellular” competence without literally modelling every cell.

4. Cost, funding, and feasibility

For the sort of thing you’re gesturing at (truly massive, multi-scale AI mind):

Training and infrastructure costs

- Cutting-edge LLM training runs are already in the tens to hundreds of millions of dollars; future frontier systems with 10–100x more compute could push into multi-billion territory.[2][11][12]
- Whole-brain and whole-organism datasets alone are forecast at ~\$100M for sophisticated sub-million-neuron models, ~\$1B for mouse-scale emulations, and tens of billions for an initial human emulation attempt.[4][12]

Who funds it?

- Realistically: coalitions of big tech, national labs, and possibly defense and biomedical consortia.
- Given projections that AI training spend could reach hundreds of billions to a trillion dollars by 2030, such projects may be justifiable as a small fraction of total AI R&D.[11][12]

So, at full ambition, you are talking about “flagship collider-scale” projects: decades, thousands of researchers, and tens of billions in capital.

5. Where 1AI and –1AI sit in the lifecycle

If we treat 1AI and –1AI as your paired advanced systems (e.g., constructive vs adversarial / stabilizing vs destabilizing):

Planning & concept phase

- You’d be here now: specifying capabilities (omnicellular reasoning, macro-simulation), safety properties, and interfaces between 1AI and –1AI.
- This is pre-prototype but deeply tied to safety and governance design.

Architectural and simulation framework design

- Next step is designing the multi-scale architecture and the sandbox worlds they’ll operate in.
- 1AI might be optimized for cooperative, high-fidelity prediction; –1AI for red-teaming, adversarial stress-testing, and identifying failure modes.

Implementation and scaling phase

- Only after early prototypes and environment simulators exist do you start throwing frontier-scale compute at these designs.
- That puts true 1AI/–1AI, at the level you’re imagining, in a post-current-frontier era—building on at least one or two more generations of scaling and infrastructure.

So in a realistic development process, 1AI/–1AI as you describe would be late-stage systems built after we have mature multi-scale simulators, robust alignment techniques, and much cheaper exascale+ compute.

6. Problems and pushback

You'd face issues on multiple fronts:

Technical problems

- Representational explosion: naïve per-cell modelling is intractable; you need abstractions and careful compression.
- Training signal: it's hard to obtain ground truth at fine granularity for large biological/physical systems, leading to epistemic uncertainty.
- Reliability and alignment: large, powerful, world-modelling AIs are harder to verify and control, especially if they can design or optimize real-world interventions.[10]

Scientific and philosophical pushback

- Neuroscience and biology communities will dispute what “fidelity” is required: do you need connectomes, detailed electrophysiology, or just behavior-level equivalence?[4][13]
- There will be arguments that you can get AGI-level cognition without massive cellular detail—so why pursue omnicellular fidelity at all?

Ethical, legal, and political resistance

- Fears about creating entities with moral status (if sufficiently brainlike) or extremely capable planning systems.
- Concerns about military use, surveillance, and centralization of power when only a few actors can afford such systems.[11][12]
- Environmental and economic concerns about devoting enormous energy and capital to training and running them.

Regulatory friction

- Expect emerging AI regulation to heavily scrutinize projects with enormous compute budgets or brain-emulation ambitions, possibly imposing caps or mandatory oversight.[10][12]
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Next Steps

If you want, we can next sketch a more concrete “tech spec” for an OMNICELLULAR ARCHITECTURE in terms of: parameter counts, memory layout, modules, and what 1AI vs –1AI each specialize in.

Citations

[1] Scaling Law Of Language Models | Towards Data Science <https://towardsdatascience.com/scaling-law-of-language-models-5759de7f830c/> [2] How Scaling Laws Drive Smarter, More Powerful AI - NVIDIA

Blog <https://blogs.nvidia.com/blog/ai-scaling-laws/> [3] Scaling Laws of Large Language Models - Emergent Mind <https://www.emergentmind.com/topics/scaling-laws-of-large-language-models> [4] Building Brains on a Computer - Asimov Press <https://press.asimov.com/articles/brains> [5] Artificial intelligence by mimicking natural intelligence <https://www.neuroai.science/p/connectomics-behavioural-cloning> [6] AI Model Size vs Performance 2025: Parameter Guide <https://localaimaster.com/blog/ai-model-size-vs-performance-analysis-2025> [7] Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture ... <https://machinelearning.apple.com/research/parameters-flops-scaling> [8] [2001.08361] Scaling Laws for Neural Language Models - arXiv <https://arxiv.org/abs/2001.08361> [9] Scaling laws - Glenn K. Lockwood <https://www.glennklockwood.com/garden/scaling-laws> [10] 2.4: Scaling Laws | AI Safety, Ethics, and Society Textbook <https://www.aisafetybook.com/textbook/scaling-laws> [11] How many AI models will exceed compute thresholds? | Epoch AI <https://epoch.ai/blog/model-counts-compute-thresholds> [12] A reason to hedge on whole-brain emulation for AGI <https://carboncopies.org/Blog/Posts/HedgeOnWholeBrainEmulation/Post/> [13] [PDF] Monte Carlo model of brain emulation development <http://www.aleph.se/papers/Monte%20Carlo%20model%20of%20brain%20emulation%20development.pdf> [14] Scaling Laws for LLMs: From GPT-3 to o3 - Deep (Learning) Focus <https://cameronrwolfe.substack.com/p/llm-scaling-laws> [15] LLM Model Size: Comparison Chart & Performance Guide (2025) <https://labelyourdata.com/articles/llm-fine-tuning/llm-model-size>